

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W009
Well Name	SYN-Assam-Duliajan-009
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.396274 N
Longitude	95.396813 E
Well Status	Completed
Well Role	Historical (offset reference well)
Spud Date	17-Dec-2009
Drilling Completed	27-Dec-2009
Report / Completion Date	04-Jan-2010
Total Depth (Driller)	3,180 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-11
Drilling Days	11 days
Recorded Events	6
Recorded NPT	57.0 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
 Well ID: W009
 Well Name: SYN-Assam-Duliajan-009
 Field: SYN-Assam-Duliajan
 Latitude: 27.396274
 Longitude: 95.396813
 Total Depth: 3,180 m
 Primary Formation: Sandstone
 Well Status: Completed
 Spud Date: 2009-12-17
 Event Count: 6
 Total NPT: 57.0 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

DOC No. WCR-W009-R0 - ISSUE 04-Jan-2010 - PROTOTYPE TRAINING RECORD

3. WELL SUMMARY

3.1 Objective

Well W009 (SYN-Assam-Duliajan-009) was drilled to drill a development location targeting the primary sandstone objective and obtain a complete formation and event record for offset reference. The well was drilled vertically from surface to a driller's total depth of 3,180 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,180 m, completed in 11 rotary drilling days between 17-Dec-2009 and 27-Dec-2009. Average overall penetration rate for the well was 12.0 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–743 m), Shale (743–1,783 m), Sandstone (1,783–2,817 m), Limestone (2,817–3,180 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 6 reportable events were recorded, comprising 2 × Cementing Issue, 1 × Wellbore Instability, 3 × Mud Loss. Aggregate non-productive time was 57.0 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 745 m, 9-5/8" at 1,785 m, 7" at 3,180 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Completed**. The well was completed on the primary sandstone objective and handed over for production testing.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W004	SYN-Assam-Duliajan-004	5.02	3,590	Sandstone	2
W006	SYN-Assam-Duliajan-006	6.18	3,960	Sandstone	0
W001	SYN-Assam-Duliajan-001	7.01	3,470	Sandstone	4
W003	SYN-Assam-Duliajan-003	9.95	3,850	Sandstone	8
W002	SYN-Assam-Duliajan-002	10.71	3,450	Sandstone	7
W007	SYN-Assam-Duliajan-007	10.80	3,400	Sandstone	3

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	743	743	Cementing Issue recorded (2 events)
2	Shale	743	1,783	1,040	Routine drilling, no reportable event
3	Sandstone	1,783	2,817	1,034	Wellbore Instability, Mud Loss recorded (4 events)
4	Limestone	2,817	3,180	363	Routine drilling, no reportable event

Formation: Claystone | Top: 0 m | Bottom: 743 m
 Formation: Shale | Top: 743 m | Bottom: 1783 m
 Formation: Sandstone | Top: 1783 m | Bottom: 2817 m
 Formation: Limestone | Top: 2817 m | Bottom: 3180 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–743	9.56	9.1–9.9	32.6	137	8.3
Shale	743–1,783	10.86	10.3–11.4	16.3	127	16.3
Sandstone	1,783–2,817	12.12	11.4–12.9	15.8	117	22.9
Limestone	2,817–3,180	13.51	13.4–13.6	8.0	98	32.6

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 13.5 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 123 rpm range on average with weight on bit averaging 18.4 klbf. Penetration rate declined with depth from 32.6 m/hr in the upper hole to 8.0 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0053	500	Claystone	Cementing Issue	Medium	7.4
E0052	592	Claystone	Cementing Issue	Low	3.7
E0056	1,879	Sandstone	Wellbore Instability	High	19.0
E0054	1,921	Sandstone	Mud Loss	High	15.3
E0051	2,068	Sandstone	Mud Loss	Medium	5.5
E0055	2,597	Sandstone	Mud Loss	Medium	6.1

Well ID: W009 | Event Type: Cementing Issue | Depth: 500 m | Formation: Claystone | Severity: Medium | Duration: 7.4 hours
 Well ID: W009 | Event Type: Cementing Issue | Depth: 592 m | Formation: Claystone | Severity: Low | Duration: 3.7 hours
 Well ID: W009 | Event Type: Wellbore Instability | Depth: 1879 m | Formation: Sandstone | Severity: High | Duration: 19.0 hours
 Well ID: W009 | Event Type: Mud Loss | Depth: 1921 m | Formation: Sandstone | Severity: High | Duration: 15.3 hours
 Well ID: W009 | Event Type: Mud Loss | Depth: 2068 m | Formation: Sandstone | Severity: Medium | Duration: 5.5 hours
 Well ID: W009 | Event Type: Mud Loss | Depth: 2597 m | Formation: Sandstone | Severity: Medium | Duration: 6.1 hours

7.1 EVENT DETAIL (1-3 OF 6)

EVENT E0053 — CEMENTING ISSUE AT 500 m (CLAYSTONE)

Well ID: W009
 Event ID: E0053
 Event Type: Cementing Issue
 Depth: 500 m
 Formation: Claystone
 Severity: Medium
 Duration: 7.4 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Poor cement bond opposite Claystone at 500 m; channelling indicated on the bond log and a remedial squeeze planned. Parameters recorded around the event depth: mud weight 9.83 ppg, ROP 33.8 m/hr, 133 rpm, WOB 10.2 kbf.

Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage.

Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

EVENT E0052 — CEMENTING ISSUE AT 592 m (CLAYSTONE)

Well ID: W009
 Event ID: E0052
 Event Type: Cementing Issue
 Depth: 592 m
 Formation: Claystone
 Severity: Low
 Duration: 3.7 hours
 Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment.

Description. Poor cement bond opposite Claystone at 592 m; channelling indicated on the bond log and a remedial squeeze planned. Parameters recorded around the event depth: mud weight 9.81 ppg, ROP 28.0 m/hr, 139 rpm, WOB 10.3 kbf.

Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage.

Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

EVENT E0056 — WELLBORE INSTABILITY AT 1,879 m (SANDSTONE)

Well ID: W009
 Event ID: E0056
 Event Type: Wellbore Instability
 Depth: 1879 m
 Formation: Sandstone
 Severity: High
 Duration: 19.0 hours
 Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Borehole collapse tendency in Sandstone at 1879 m; repeated back-reaming needed to maintain hole condition. Parameters recorded around the event depth: mud weight 11.51 ppg, ROP 18.4 m/hr, 122 rpm, WOB 19.0 kbf.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cuttings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cuttings volume is the earliest indicator.

7.2 EVENT DETAIL (4-6 OF 6)

EVENT E0054 — MUD LOSS AT 1,921 m (SANDSTONE)

Well ID: W009
 Event ID: E0054
 Event Type: Mud Loss
 Depth: 1921 m
 Formation: Sandstone
 Severity: High
 Duration: 15.3 hours
 Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Loss of circulation into porous Sandstone at 1921 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 11.51 ppg, ROP 18.4 m/hr, 122 rpm, WOB 19.0 kbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0051 — MUD LOSS AT 2,068 m (SANDSTONE)

Well ID: W009
 Event ID: E0051
 Event Type: Mud Loss
 Depth: 2068 m
 Formation: Sandstone
 Severity: Medium
 Duration: 5.5 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Loss of circulation into porous Sandstone at 2068 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 11.85 ppg, ROP 15.8 m/hr, 117 rpm, WOB 21.1 kbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0055 — MUD LOSS AT 2,597 m (SANDSTONE)

Well ID: W009
 Event ID: E0055
 Event Type: Mud Loss
 Depth: 2597 m
 Formation: Sandstone
 Severity: Medium
 Duration: 6.1 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Loss of circulation into porous Sandstone at 2597 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 12.42 ppg, ROP 17.4 m/hr, 106 rpm, WOB 23.7 kbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	745	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	1,785	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 630 m
7"	N-80, 29 lb/ft	3,180	8-1/2"	Tail 15.8 ppg, TOC 1,640 m

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0053** — Cementing issue at 500 m in Claystone (Medium, 7.4 hr). Poor cement bond opposite Claystone at 500 m; channelling indicated on the bond log and a remedial squeeze planned.
- **E0052** — Cementing issue at 592 m in Claystone (Low, 3.7 hr). Poor cement bond opposite Claystone at 592 m; channelling indicated on the bond log and a remedial squeeze planned.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,180 m
Primary Formation	Sandstone
Major Events	2 of 6 events rated High or Critical
Total NPT	57.0 hours
Completion Status	Completed
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Cementing Issue** at 500 m, 592 m (Claystone) — 11.1 hr cumulative.
- **Wellbore Instability** at 1,879 m (Sandstone) — 19.0 hr cumulative.
- **Mud Loss** at 1,921 m, 2,068 m, 2,597 m (Sandstone) — 26.9 hr cumulative.

9.2 Key Historical Observations

- Annular coverage is sensitive to losses and centralisation across the shoe interval.
- Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.
- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Review slurry design, centralisation and displacement rate for the stage covering 500 m; run a bond log before proceeding.
- Raise inhibition before entering the Sandstone section and limit open-hole exposure time; treat rising cavings volume as an early warning.
- Approach the Sandstone interval (1,921–2,597 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.

10. DOCUMENT CONTROL

Document Number	WCR-W009-R0
Revision	R0
Issue Date	04-Jan-2010
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formation.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.